EMOTION RECOGNITION FROM SOCIAL MEDIA TEXT USING LARGE LANGUAGE MODELS

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Abstract — Emotion recognition from social media text is challenging because short online messages frequently contain informal language, emojis, slang, implicit meaning, sarcasm, and context-dependent expressions. This paper presents a deep-neural and pragmatic framework for identifying emotions in user-generated text. The proposed design combines contextual representations from a transformer-based language model with a lightweight pragmatic layer that considers linguistic cues such as negation, idiomatic expressions, and incongruity. The pipeline includes text validation and preprocessing, contextual emotion scoring, pragmatic adjustment, decision calibration, and presentation of the predicted label. A multi-label emotion dataset such as GoEmotions can support training and evaluation, provided the selected label mapping and data split are documented. Performance should be measured using precision, recall, F1-score, and suitable multi-label metrics; confusion analysis should separately examine sarcasm, ambiguity, and class imbalance. The framework is intended to make emotion classification more context-aware than keyword-only approaches while retaining a modular design that can be inspected and improved. Quantitative performance claims are intentionally not reported here because verified experimental measurements were not supplied with the project materials.

Keywords—Emotion recognition, social media text, natural language processing, transformers, RoBERTa, GoEmotions, pragmatic reasoning, calibration.

I. INTRODUCTION

Social media platforms produce large volumes of short text in posts, comments, replies, and reviews. These messages can express joy, sadness, anger, fear, love, surprise, and other emotional states, often without using explicit emotion words. Emotion recognition differs from coarse sentiment analysis: sentiment typically describes polarity, whereas emotion classification aims to identify a more specific affective category [1].

The task is difficult because social media language is noisy and context-sensitive. For example, emojis may change the tone of a sentence, negation can reverse an apparent meaning, and sarcasm may communicate an emotion that conflicts with the literal wording. Traditional bag-of-words methods provide useful baselines, but they do not explicitly model contextual relationships. Transformer models offer contextual representations that can support more nuanced classification [2], [3].

This paper describes a modular emotion intelligence framework that combines a transformer-based neural classifier with pragmatic cues and calibration. Its intended contribution is not merely to assign a label, but to provide a structure in which contextual prediction, rule-based adjustments, and decision thresholds can be evaluated independently. The remaining sections review related work, explain the framework, define evaluation requirements, and discuss limitations and future extensions.

II. RELATED WORK

Traditional text classification commonly represents documents using bag-of-words or TF–IDF features and applies classifiers such as logistic regression, naïve Bayes, or support vector machines. These methods are computationally efficient and useful as baselines, but their features may fail to capture long-distance context and implicit emotional meaning.

Deep neural approaches, including convolutional and recurrent networks, learn representations from labelled examples. Transformer architectures improve contextual modelling through self-attention. BERT introduced bidirectional transformer pretraining for language understanding [2], while RoBERTa refined the pretraining recipe and demonstrated strong results on language understanding tasks [3]. These models can be adapted to emotion classification by fine-tuning on a labelled dataset.

Demszky et al. introduced GoEmotions, a dataset of approximately 58,000 Reddit comments annotated with 27 emotion categories and a neutral category [1]. Its fine-grained labels make it useful for studying emotion recognition, although its source domain and multi-label annotations must be considered when transferring a model to other social platforms. Research on sarcasm detection also shows that literal lexical content may conflict with intended meaning, making context important [4].

The framework described here addresses this practical gap by keeping neural prediction and pragmatic adjustment as

separate components. This modularity allows the effect of pragmatic rules and calibration to be measured against a neural-only baseline rather than assuming that additional rules always improve performance.

III. PROPOSED METHODOLOGY

A. System Overview

The processing pipeline is: input text validation → conservative preprocessing → contextual model inference → pragmatic cue analysis → calibrated decision → output label and confidence information. Each stage should log its configuration so that predictions can be reproduced. The implementation may expose the pipeline through a lightweight web interface or API.

B. Text Preparation

Input text is checked for missing or empty content and normalised only where necessary. Duplicate records and invalid examples are handled during dataset preparation. URLs and user mentions may be replaced with stable tokens, while emotionally informative emojis, hashtags, punctuation, and negation should not be removed without testing their impact. Excessive cleaning can erase cues needed to infer emotion.

C. Contextual Emotion Model

A transformer encoder such as RoBERTa can be fine-tuned on labelled emotion examples. The model converts the tokenised input into contextual representations and produces scores for the supported emotion labels. If the chosen dataset uses multi-label annotations, the output layer and loss function should support multiple simultaneous labels; if the application requires one label, the label-selection policy must be stated explicitly. The exact checkpoint, tokenizer, training configuration, label mapping, and split sizes must be recorded in the final experimental version.

D. Pragmatic Analysis

The pragmatic layer examines cues that may be poorly represented by literal lexical meaning alone. Candidate signals include negation, known idioms, contrastive constructions, and potential incongruity between a phrase and its surrounding context. Rules should be conservative, explainable, and tested individually. They may adjust or flag model scores, but should not override the neural prediction indiscriminately. Rule coverage and false adjustments should be recorded during error analysis.

E. Calibration and Decision Policy

Raw model scores are not automatically reliable probabilities. Calibration can be assessed on a validation set, and per-class thresholds can be selected when label prevalence or error costs differ across emotions. Thresholds must be chosen using validation data only and then frozen before final testing. A separate test set is needed to avoid reporting optimistic estimates caused by tuning on test examples.

F. Experimental Protocol

A reproducible evaluation should compare at least (1) a neural-only baseline, (2) the neural model with pragmatic adjustments, and (3) the full model with calibration and class-specific decision thresholds. Use the same data splits for all variants. For multi-label classification, report micro- and macro-F1 and per-class precision and recall where appropriate; for single-label classification, report accuracy and macro-F1 in addition to the confusion matrix. Also measure inference latency and inspect errors involving sarcasm, negation, ambiguous phrasing, and underrepresented classes.

IV. RESULTS AND EVALUATION PLAN

The project documentation identifies accuracy, precision, recall, F1-score, and confusion-matrix analysis as appropriate evaluation methods. However, the supplied materials do not provide verified dataset counts, model checkpoint details, completed test metrics, or measured latency for the proposed framework. Therefore, this paper does not claim a numerical improvement or include invented performance values. The final version should replace the table below with results obtained by running the documented experiment.

Approach	Accuracy	Precision	Recall	F1-score
Neural-only baseline	To measure	To measure	To measure	To measure
Neural + pragmatic layer	To measure	To measure	To measure	To measure
Full model + calibration	To measure	To measure	To measure	To measure

Results should be interpreted comparatively. If pragmatic adjustments improve recall for selected emotion classes but reduce precision elsewhere, that trade-off should be reported rather than summarised as a universal improvement. Calibration should be assessed separately from classification quality, and latency should be measured on a stated hardware and software environment. A confusion matrix and a small, privacy-safe set of representative error cases can help explain the measured outcomes.

V. DISCUSSION

The principal design choice is to separate contextual neural inference from explicit pragmatic reasoning. This makes the system easier to inspect and allows ablation experiments to determine whether each component adds value. A transformer model provides broad contextual capability, while a rule layer can encode selected linguistic patterns that are important to the target application. Calibration provides a further mechanism for adjusting decision behaviour when scores or class distributions are uneven.

These benefits remain hypotheses until validated experimentally. Hand-written rules can be brittle, language- and domain-specific, and may misinterpret figurative language. Fine-grained emotion labels can overlap, and annotations may reflect annotator disagreement. A model trained on Reddit text may not generalise to another platform, language, or population. Privacy, bias, and the

responsible handling of user-generated text must also be considered.

VI. CONCLUSION AND FUTURE WORK

This paper presents a modular framework for emotion recognition from social media text using contextual neural modelling, pragmatic cues, and calibration. The design covers text preparation, transformer inference, rule-based analysis, decision thresholds, and a reproducible evaluation protocol. The available project materials do not contain completed measurements, so quantitative effectiveness cannot yet be concluded. A valid final assessment requires running the experiments and reporting the actual test results.

Future work includes expanding evaluation to multilingual and code-mixed text, improving sarcasm and idiom handling, testing explainable prediction summaries, measuring robustness across domains, and evaluating bias across emotion classes. Deployment work can also examine real-time inference, latency, and privacy-preserving processing.

ACKNOWLEDGMENT

The author acknowledges the guidance and academic support provided by the Department of Computer Science and Engineering at Koneru Lakshmaiah Education Foundation. The final submitted version should update this statement to reflect the actual contributors and approvals.

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